

# Internship Final Report

## Inter Information

Student Name: Gopichand Dandimeni

University: Sandip university

Major: B-tech specializes in Cybersecurity & Digital Forensics

Internship Duration: July 1st, 2025 - July 31st, 2025

Company: ShadowFox

Domain: Cyber Security

Mentor: Mr. Pranshu

Assistant Mentor: Mr. Pranshu

Coordinator: Mr. Aakash

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## Objectives

### The objectives of the internship are as follows:

- To impart foundational to advanced level practical skills in cybersecurity through hands-on tasks.
- To appreciate offensive security techniques, such as reconnaissance techniques, brute-forcing, traffic analysis, cryptography, reverse engineering, and various types of network attacks.
- To maintain ethical hacking practices by exploiting chosen testing environments, such as testphp.vulnweb.com or the TryHackMe room.
- To enforce Proof of Work (POW) by having the verifications of the task by mentors prior to submission, assuring original and honest work.

## Tasks and Responsibilities

### Pre-Internship Requirements:

- The Updated LinkedIn Profile should include "ShadowFox Cyber Security Intern" and relevant skills.

### Tasks:-

#### ❖ Beginner:

- I. Port Scan on testphp.vulnweb.com for opened ports.
- II. Brute-forcing to find directories on the testphp.vulnweb.
- III. Network sniffing with Wireshark when logged in to capture the credentials.

### ❖ Intermediate:

- I. Crack the hash from encoded.txt to reveal a secret code by decrypting the VeraCrypt file.
- II. Analyze a VeraCrypt Executable file using a PE explorer to locate the entry point address.
- III. Generate a Metasploit payload for another OS in VMWARE to establish a reverse shell connection.

### ❖ Hard Level

- I. Completing TryHackMe's "Basic Pentesting" room, answering all questions and documenting the attack methodology.

### Responsibilities:

- Maintain confidentiality (e.g., no sharing of reports with other members).
- Submit validated Proof of Work via mentor approval before task submission.
- I will attend all the sessions conducted by the mentor and the team.
- I will communicate regularly with my mentor throughout the internship.

### Learning Outcomes

- I have undergone practical training in the cyber security field with hands-on practices for port scanning, brute force attacks, and network traffic interception.
- I have acquired skills in using tools such as Wireshark, Metasploit, and PE-Explorer for penetration testing and network analyses.
- I improved my understanding of encryption techniques and file decryption using Veracrypt.
- I enhanced my problem-solving capabilities by addressing real cybersecurity problems and finding suitable solutions.
- I gained some insight into the significance of Proof of Work and the validation process within a professional setup.

### Challenges and Solutions

- The challenge was to understand the network traffic that was captured during the interception task.  
**Solution:** Advanced studies were undertaken via online resources and tutorials to learn more about Wireshark functions and analyze sample traffic to enhance understanding.
- Challenge: Problems were encountered while performing a payload using Metasploit because of unfamiliarity with the tool.  
**Solution:** Following step-by-step tutorials and interacting with online forums helped troubleshoot and create the payload.
- Challenge: Cracking the Wi-Fi password during the deauthentication attack proved to be more difficult than expected.  
**Solution:** A comprehensive wordlist was built by looking into commonly used password orientations while utilizing multiple techniques of cracking to increase the chances of success.
- The all in one solutions for all my challenges I faced in this intership I ued "Blackbox AI"

## **Conclusion**

The internship was a superb opportunity to apply theoretical knowledge in practice. By performing a wide range of tasks, I was able to sharpen my technical skills and better understand the nuances inherent in cybersecurity practice. The challenges encountered during the internship were key factors in training my problem-solving abilities and cultivating resilience. Overall, this experience convinced me to make cyber security my career of choice, enabling me to acquire skills that will be vital for the pursuit of my future endeavors.

## **Acknowledgments**

Special thanks to my mentor and the ShadowFox team, whose guidance and support throughout my internship were extremely helpful. Their invaluable advice assisted me with the various tasks and challenges that were assigned to me. I would also like to thank you for equipping me with the resources and tools that greatly impacted my learning experience. This is a wonderful opportunity to learn and gain experience in cybersecurity.